

Founding AI Engineer — Assignment

Assignment: AI-Powered Frontend Website Cloning Agent

Objective

Build an **AI Agent** that takes a publicly accessible website URL and automatically recreates its **frontend/UI**.

The goal is to evaluate your ability to build an AI-powered engineering system that can analyze an unfamiliar website, generate a frontend, and modify it using natural-language instructions.

1. Expected Workflow

```
Website URL
  ↓
AI Agent
  ↓
Analyze Website
  ↓
Understand UI / Layout
  ↓
Generate React / Next.js Frontend
  ↓
Run & Validate
  ↓
Local Preview
  ↓
Modify using AI Prompts
```

The output should be a **new frontend implementation**, not an embedded copy of the original website.

2. Core Requirements

The AI Agent should:

- Accept a public website URL.
 - Analyze the website's:
 - Layout
 - Sections
 - Navigation
 - Text
 - Images/assets
 - Colors
 - Typography
 - Spacing
 - Components
 - Responsive structure
 - Generate a responsive **React/Next.js** frontend.
 - Recreate the original UI as accurately as reasonably possible.
 - Use reusable components and clean code.
 - Detect and handle basic generated-code/build errors.
 - Provide a local preview of the generated website.
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3. AI-Based Modification

After generating the website, the user should be able to modify it using natural-language instructions.

Examples:

| "Change the primary color to blue."

| "Add a testimonials section."

| "Replace the hero section with a bakery hero."

| "Make the navbar sticky."

| "Remove the pricing section."

The AI Agent should modify the generated frontend/code based on these instructions.

4. Multiple Website Test

Your solution should **not be hardcoded for one website**.

We will provide **3 different websites** for evaluation.

Your agent should attempt to recreate all three.

5. Technology

Preferred:

- React / Next.js
- TypeScript

You are free to choose other technologies where appropriate.

You may use AI development tools such as:

- ChatGPT
- Claude
- Cursor
- GitHub Copilot
- Other AI coding tools

Using AI tools is encouraged. You should, however, be able to explain your implementation and architecture.

6. Hosting

No hosting is required.

The entire assignment only needs to **run locally**.

7. Submission

1. Complete Codebase

GitHub repository containing the working project.

2. Demo Video

A **5–10 minute video** showing:

1. Entering a website URL
2. Website analysis
3. AI Agent generating the frontend
4. Generated website preview
5. Responsive/mobile view
6. Modifying the website using an AI prompt

3. README

Keep the README concise and include:

- Setup instructions
- Architecture
- Technologies/models used
- Key implementation decisions
- Limitations

4. Architecture Diagram

Show how your AI Agent works from:

URL → Analysis → Generation → Validation → Modification

Evaluation

Area	Weight
Frontend recreation quality	25%
AI Agent implementation	20%
Generalization across websites	20%
Code quality & architecture	15%
Natural-language modification	10%
Error handling	5%
Cost awareness	5%
Total	100%

Technical Discussion

After submission, we may discuss:

- Why you chose your architecture
- How your agent analyzes a website
- How you generate reliable frontend code
- How you handle generated-code errors
- How you improve visual accuracy
- How you would reduce AI/API costs
- How you would scale the system
- What you would improve with more development time

Time Limit

48 hours

We are **not expecting a production-ready platform**.

We are looking for a strong working MVP that demonstrates your ability to **build AI agents, work with unfamiliar systems, develop frontend applications quickly,**

and solve problems independently.